Ramp energy lab

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Compare gravitational energy change with measured kinetic energy.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 10.*
K=12 J.

02. Worked example: explain the model
U loss 6=translational gain 4+other 2; do not automatically call every difference measurement error.

03. Read and interpret the model
Vertical height.

04. Apply the physics
Lost U 5 J; difference 1 J.

05. Explain your reasoning
Teacher checks measured data, units and clear reference; no invented results.

06. Review: connect an earlier idea *Review: Unit 4, Day 1.*
Factor 16, because $K=\frac{1}{2}mv^2$.

07. Exit check
Rotational energy and thermal energy; others justified.

08. STRETCH: test the model
Measure rotation or compare a controlled sliding setup, keep heights/masses consistent and track uncertainties.

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01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 10.*
$h=v_i^2/(2\ g)=11.25$ m.

02. Worked example: explain the model
U loss 6=translational gain 4+other 2; do not automatically call every difference measurement error.

03. Read and interpret the model
Vertical height.

04. Apply the physics
Initial U=0.98 J; translational K 1.5625 J, too large for assumptions. Check push, height, mass, speed calibration/location and uncertainty.

05. Explain your reasoning
Teacher checks measured data, units and clear reference; no invented results.

06. Review: connect an earlier idea *Review: Unit 4, Day 1.*
K becomes 1/4 of the original; fixed p means $K \propto 1/m$.

07. Exit check
1.96 J; 1.44 J; 73.5%.

08. STRETCH: test the model
U drop 2 J; K 1 J; remainder 1 J. Conditional friction magnitude 1 N; rotation would alter that attribution.